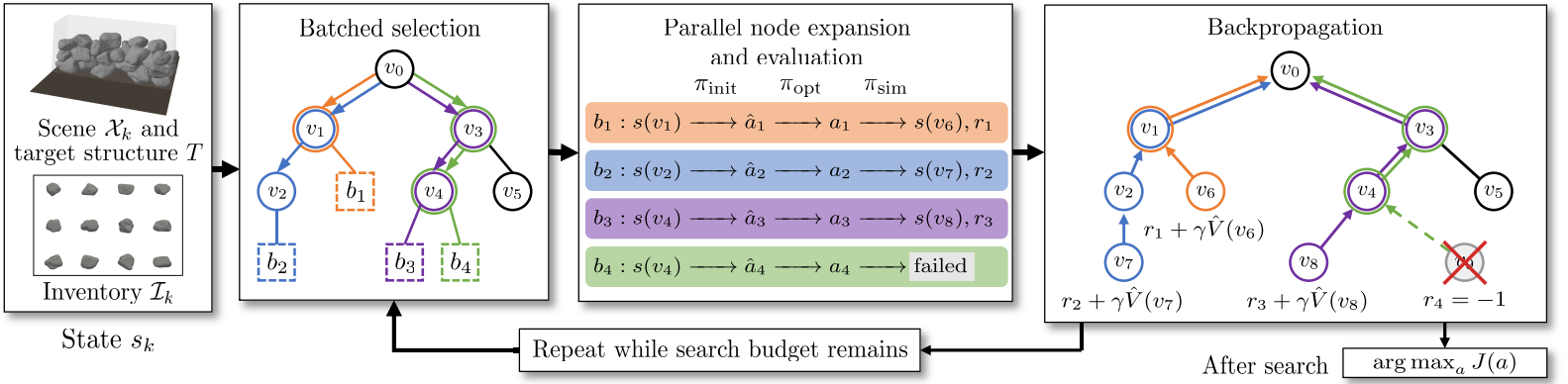

Scene $\mathcal{X}_k$ and target structure T
Inventory $\mathcal{I}_k$
State s_k
Batched selection
v_0
v_1
v_2
b_1
v_3
v_4
v_5
b_2
b_3
b_4
Parallel node expansion and evaluation
π_{init} π_{opt} π_{sim}
$b_1 : s(v_1) \longrightarrow \hat{a}_1 \longrightarrow a_1 \longrightarrow s(v_6), r_1$
$b_2 : s(v_2) \longrightarrow \hat{a}_2 \longrightarrow a_2 \longrightarrow s(v_7), r_2$
$b_3 : s(v_4) \longrightarrow \hat{a}_3 \longrightarrow a_3 \longrightarrow s(v_8), r_3$
$b_4 : s(v_4) \longrightarrow \hat{a}_4 \longrightarrow a_4 \longrightarrow$ failed
Backpropagation
v_0
v_1
v_3
v_2
v_6
v_4
v_5
v_7
v_8
v_9
$r_1 + \gamma \hat{V}(v_6)$
$r_2 + \gamma \hat{V}(v_7)$
$r_3 + \gamma \hat{V}(v_8)$
$r_4 = -1$
Repeat while search budget remains
After search
$\arg\max_a J(a)$